Lena Collienne

Curriculum Vitae

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github.com/lenacoll

Research

Lecturer, *University of Otago*.
Developing deep learning methods for phylogenetics.

Postdoctoral Research, *Fred Hutchinson Cancer Center*,
(Advisor: Frederick Matsen IV, PhD).
Developing novel methods for phylogenetic inference using machine learning.

Postdoctoral Research, *University of Canterbury*,
(Advisor: Alex Gavryushkin, PhD).
Extending subtree prune and regraft operations to ranked phylogenetic trees and investigating thereby introduced distance measures for phylogenetic time trees.

PhD Research, *University of Otago*,
(Advisors: Alex Garvyushkin, PhD, and David Bryant, PhD).
Introducing and analysing spaces of phylogenetic time trees based on nearest neighbour interchange tree rearrangements.

M.Sc. Research, *University of Greifswald*,
(Advisors: Mareike Fischer, PhD, and Alex Gavryushkin, PhD).
Establishing properties of ranked nearest neighbour interchange moves between ranked phylogenetic trees

Education

2018–2021 **Doctor of Philosophy**, *Computer Science*, University of Otago (NZ).
2016–2018 **Master of Science**, *Biomathematics*, University of Greifswald (GER).
2012–2015 **Bachelor of Science**, *Biomathematics*, University of Greifswald (GER).

Work Experience

2026–present **Lecturer**, *School of Computing*, University of Otago (NZ).
2023–2026 **Postdoctoral Research Fellow**, *Matsen group*, Fred Hutch Cancer Center (US).
2022–2023 **Postdoctoral Research Fellow**, *BioDS lab, School of Mathematics and Statistics*, University of Canterbury (NZ).

Teaching Experience

2023–2024 **Facilitator**, *Girls Who Code*, Fred Hutch Cancer Center (Seattle, WA, USA).
2022 **Lecturer**, *STAT211: Random Processes*, University of Canterbury (NZ).

2019 **Tutor**, *COSC341: Theory of Computing*, University of Otago (NZ).

2015–2016 **Summer Research Project**, *University of Auckland (NZ)*.

Scholarships and Awards

2022 *Hatherton Award* (Royal Society of New Zealand)

2021 *Exceptional PhD thesis* (Division of Science, University of Otago)

2018–2021 *University of Otago Doctoral Scholarship*

2018 *Externally Funded Research Grant* (Max Planck Institute Plön)

2015 *Summer Research Scholarship* (University of Auckland)

2015 *PROMOS Travel Scholarship* (University of Greifswald)

2014–2015 *Deutschlandstipendium* (Alfried Krupp von Bohlen und Halbach Foundation/Federal Government of Germany)

Additional Activities

2021 President of the Otago Computer Science Society (University of Otgao)

2019–2021 Member of the Postgraduate Committee (Department of Computer Science, University of Otago)

2019–2021 Organising the annual Postgraduate Symposium (Department of Computer Science, University of Otago)

2015–2018 Member of Student Council (Institute for Mathematics and Computer Science, University of Greifswald)

Talks

Conference Talks

2026 **Uniting Phylogenetic Network Research**, *Lorentz Center, Leiden (NED)*.
Invited talk: Deep learning for phylogenetics? Learning from classical algorithms

2024 **Evolution 2024**, *Montreal (CA)*.
Contributed talk: Instability in Phylogenetic Trees after Taxon Addition

2023 **SMB meeting 2023**, *Columbus (OH, US)*.
Invited minisymposium talk: Spaces of Discrete Time Trees

2022 **Phylomania 2022**, *Hobart (AU)*.
Contributed talk: Subtree Prune and Regraft on Ranked Trees

2021 **Phylomania 2021** (Best Student Talk Award), *Online*.
Contributed talk: Distances between Phylogenetic Time Trees

2021 **NZ Phylogenomics Meeting**, *Akaroa (NZ)*.
Contributed talk: The Space of Discrete Coalescent Trees

2020 **NZ Phylogenomics Meeting**, *Waiheke (NZ)*.
Contributed talk: Online Algorithms in Computational Biology

2019 **NZ Phylogenomics Meeting**, *Napier (NZ)*.
Contributed talk: The Ranked Nearest Neighbour Interchange Space of Phylogenetic Trees

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Invited Seminar Talks

2026 **Centre for Artificial Intelligence in Public Health Research**, *Robert Koch Institute (GER)*.
Deep Learning for Phylogenetic Inference

2025 **Department of Mathematics and Computer Science**, *University of Greifswald (GER)*.
Using Deep Learning to Guide Phylogenetic Tree Search

2021 **Departmant of Mathematics**, *University of Otago (NZ)*.
The Space of Discrete Coalescent Trees

2020 **Online Seminars on Algorithms and Complexity in Phylogenetics**, *Online*.
Computing the Ranked Nearest Neighbour Interchange Distance between Ranked Phylogenetic Trees

2019 **Max Planck Institute for Mathematics in the Science**, *Leipzig (GER)*.
The Ranked Nearest Neighbour Interchange space of phylogenetic trees

2017 **Computational Evolution Group**, *ETH Zurich (CH)*.
Discrete Time Trees

Poster Presentations

2024 **ICERM workshop "Algorithmic Advances and Implementation Challenges: Developing Practical Tools for Phylogenetic Inference"**, *Providence (RI, US)*.
Poster: Instability in Phylogenetic Trees after Taxon Addition

2022 **Phylomania 2022**, *Hobart (AU)*.
Poster: Subtree Prune and Regraft on Ranked Trees

Other

2025 **Association for Women in Mathematics**, *University of Texas (US)*.
An Unexpected Journey

2022 **School of Mathematics and Statistics**, *University of Canterbury (NZ)*.
How to Give a (Good) Talk

2020 **Postgraduate Symposium** (1st place Best Presentation Award), *University of Otago (NZ)*.
The Complexity of Computing the RNNI Distance between Phylogenetic Trees

2020 **Seminar of Departments of Computer Science and Information Science**, *University of Otago (NZ)*.
The Complexity of Computing Nearest Neighbour Interchange Distances between Ranked Phylogenetic Trees

2019 **Postgraduate Symposium** (2nd place Best Presentation Award), *University of Otago (NZ)*.
Online Algorithms in Computational Biology

Publications

Collienne, L. & Matsen IV, FA. (2026). Tree rearrangement graphs admit paths of decreasing Robinson-Foulds distance *Bulletin of Mathematical Biology*. *https://doi.org/10.1007/s11538-026-01748-7*

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Collienne, L., Richman, H., Rich, DH., Barker, M., Jennings-Shaffer C., & Matsen IV, FA. (2026). Unifying phylogenetic traversal and deep learning to guide tree exploration *Systematic Biology. https://doi.org/10.1093/sysbio/syag062*

Collienne, L., Barker, M., Suchard, MA., & Matsen IV, FA. (2024). Phylogenetic tree instability after taxon addition: empirical frequency, predictability, and consequences for online inference. *Systematic Biology syae059. https://doi.org/10.1093/sysbio/syae059*

Collienne, L., Whidden, C. & Gavryushkin, A. (2024). Ranked Subtree Prune and Regraft. *Bulleting of Mathematical Biology 86, 24. https://doi.org/10.1007/s11538-023-01244-2*

Berling, L., **Collienne, L.** & Gavryushkin, A. (2024). Estimating the mean in the space of ranked phylogenetic trees. *Bioinformatics, Volume 40, Issue 8*. https://doi.org/10.1093/bioinformatics/btae514.

Collienne, L. (2021). Spaces of phylogenetic time trees (Thesis, Doctor of Philosophy). University of Otago. Retrieved from http://hdl.handle.net/10523/12606

Collienne, L., Elmes, K., Fischer, M., Bryant, D. & Gavryushkin, A. (2021). Discrete Coalescent Trees. *Journal of Mathematical Biology 83.5, p. 60. issn: 1432-1416.*

Collienne, L. & Gavryushkin, A. (2021). Computing nearest neighbour interchange distances between ranked phylogenetic trees. *Journal of Mathematical Biology 82.1, p. 8. issn: 1432-1416*.

Preprints

Bouckaert, R., **Collienne, L.** & Gavryushkin, A. (2022). Online Bayesian Analysis with BEAST2. *BioRxiv https://www.biorxiv.org/content/10.1101/2022.05.03.490538v1*.

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